

JNEOPALLIUM · AD-FRAUD GUARDIAN

Stop Paying for Clicks That Never Happened

Multi-label invalid-traffic intelligence — names the fraud, shows the evidence, recommends the action

On-premises · Shadow → Advisory · Privacy by design · macro-F1 $\approx$ 0.97 · BSD-3 core

01 A chunk of your ad budget buys nothing

Industry estimates put invalid-traffic losses in the tens of billions of dollars a year.

- ▶ **Bots** click and view ads no human ever sees.
- ▶ **Click farms & incentivized installs** look like users but never become customers.
- ▶ **Attribution fraud** (click spam, click injection) pays the wrong partner for conversions you'd have won anyway.
- ▶ **Made-for-advertising & spoofed inventory** soak up spend behind faked supply paths.
- ▶ Every one of these quietly drains ROAS — and a single "fraud %" can't tell them apart.

02 One "fraud score" can't run your business

A single risk number

- "73% fraud" — but **which** kind?
- No evidence, so no defensible action
- Wrong action withholds a real payout
- One threshold over- or under-flags everything

What you actually need

- **Which** fraud family, named
- **Why** — the contributing evidence
- **What to do** — a proportionate action
- **Calibrated** probabilities you can trust

03 The solution: name it, prove it, recommend it

A multi-label invalid-traffic engine — not one score, ten calibrated verdicts.

- ▶ **Multi-label.** One event can be flagged as several things at once, each with its own probability.
- ▶ **Evidence-first.** Every verdict carries the signals and a human-readable reason.
- ▶ **Advisory by construction.** Candidate actions only — it never blocks a publisher or withholds a payout on its own.
- ▶ **Private by design.** Identifiers are hashed; raw IPs, location, and user-agents are never exported.
- ▶ **Explainable, on-premises, open core** — it fits review, audit, and a publisher appeal.

04 It reads every event across many clocks

Instant checks plus the slow truth that only arrives days later.

- ▶ **Instant:** forged signatures, replayed events, click-before-impression — caught on the spot.
- ▶ **Per-session:** click-injection timing, click-spam volume, attribution anomalies.
- ▶ **Minutes:** click-farm device/account fanout graphs.
- ▶ **Days:** the real truth about quality — retention, refunds, chargebacks, uninstalls.
- ▶ One engine weighs an instant signal against a multi-day quality outcome in a single verdict.

05 Ten fraud families, named

Each gets its own calibrated probability — and the right action attaches to the right problem.

- ▶ **bot · clickFarm · incentivized** — manufactured or low-value engagement
- ▶ **clickSpam · clickInjection · attributionHijack** — stealing conversion credit
- ▶ **eventSpoofing · inventorySpoofing** — forged events and faked supply paths
- ▶ **accidentalOrLowValue · unknownSuspicious** — the model says "odd, not necessarily fraud" instead of over-accusing.

06 Every verdict carries its evidence

Not a black box — an auditable decision your analysts and partners can trust.

- ▶ `probabilities`: clickInjection 0.94, attributionHijack 0.91, bot 0.03 ...
- ▶ `overallInvalidTrafficProbability`: 0.95
- ▶ `recommendation`: **HOLD_PAYOUT_CANDIDATE** (a candidate — never auto-executed)
- ▶ `reasons`: "conversion observed before click", "click injection timing anomaly"
- ▶ `mode`: **ADVISORY** — and the model never accuses a named person.

07 Proof: we measured, diagnosed, and lifted it

A deliberate model upgrade raised detection quality sharply on the reference corpus.

0.97

Macro-F1 across the
ten fraud labels

0.66 → 0.97

Before → after
the model upgrade

12 / 12

Java runtime tests
passing

ADVISORY

Safety ceiling,
never exceeded

08 Safe and private by construction

Easy to demonstrate, validate, and deploy next to billing.

- ▶ **Advisory / shadow by default.** It recommends; humans decide. It never blocks or withholds on its own.
- ▶ **Never an accusation.** It emits invalid-traffic *evidence*, not a claim that a person committed fraud.
- ▶ **Privacy first.** HMAC-hashed identifiers; no raw IP, geo, or user-agent ever exported; missing ≠ zero.
- ▶ **Safe degradation.** If the model is unavailable it falls back to transparent rules — no broken request path.
- ▶ **Honest about evidence.** Reference metrics are pipeline evidence; field accuracy is earned on your data.

09 The money math is simple

Recover wasted spend and stop paying for fraudulent conversions. Illustrative example:

- ▶ Annual ad / payout spend → **\$50,000,000**
- ▶ Invalid-traffic share (conservative) → **10%** = \$5,000,000 exposed
- ▶ Detectable, actionable share recovered → **20–40%**
- ▶ **Recovered value** → **\$1,000,000 – \$2,000,000 / year**
- ▶ Price the product at a small fraction of verified savings — the ROI is not close.

10 Land and expand — paid at every step

A low-risk path from a log export to a recurring subscription. (Indicative pricing.)

- ▶ **Offline replay.** Score an exported event log; deliver named findings, evidence, and a wasted-spend estimate. **€5–15k / assessment.**
- ▶ **Shadow pilot.** Read-only scoring beside production; compare to your chargebacks and MMP rulings. **€15–40k**, credited to year one.
- ▶ **Advisory subscription.** Operator-visible verdicts to analysts and billing review. **Per event-volume band, per site / year.**
- ▶ **Bounded enforcement.** Only after first-party validation + legal review + appeal/rollback — narrowly scoped, never silent blocking.

11 Where we fit — and where we won't overreach

We add (high value)

- Naming the exact invalid-traffic family
- Evidence for every advisory
- Calibrated, multi-label probabilities
- A safe shadow → advisory path

We won't do (by design)

- Silently block a publisher
- Auto-withhold a payout
- Accuse a named person of fraud
- Replace your MMP / SSP / billing

Send us a log export. We'll show you the leak.

Export an anonymised event log, or point us at a feed — read-only, privacy-preserving.

We return named invalid-traffic findings, the evidence, and an estimated wasted-spend impact.

Open-source demo today · paid assessment in weeks · shadow pilot next quarter.

Book a paid invalid-traffic assessment.